

The Optimize Layer: Exercises & Lifestyle Habits

What to actually do — ranked by evidence, illustrated, and honest about the gap between what's proven and what's sold.

Companion to the State-of-the-Field report · derived from [04-protocols/WHAT-TO-TRACK-SYNTHESIS.md](#)
Diagrams procedurally generated (vector, reproducible, no hallucinated anatomy)
Rule start at Tier A · treat Tier B as trends · enjoy Tier C as experiments, not protocols

TIER A — DO FIRST Large effects, converging evidence. Fitness, strength, movement, sleep, apoB, metabolic health, connection.	TIER B — WORTH IT Real but smaller / surrogate / dose-uncertain. Zone 2, sauna, mobility, breath, early-TRE, light hygiene.	TIER C — OPTIONAL Sold harder than proven. Cold plunge "for longevity", NAD ⁺ , CGM-for-healthy, age clocks.
---	---	---

The one idea behind this guide

The interventions with the largest, most replicated effects on how long and how well you live are **behavioral and unglamorous**. They are not for sale, which is exactly why they get crowded out by supplements and gadgets. This guide puts them first, illustrates the foundational movements, and labels everything by what the evidence has actually earned.

THREE RULES THAT GOVERN EVERY RECOMMENDATION HERE

(1) Predictor ≠ lever — a number that predicts death isn't automatically something that, when changed, prevents it. **(2) Cohort ≠ proof** — you can't randomize fitness or sleep over decades, so the strong-looking numbers carry bias. **(3) Something beats nothing** — the steepest gains are at the *start* of every dose-response curve; optimization past that is real but smaller than advertised.

Part 1 — The movement program

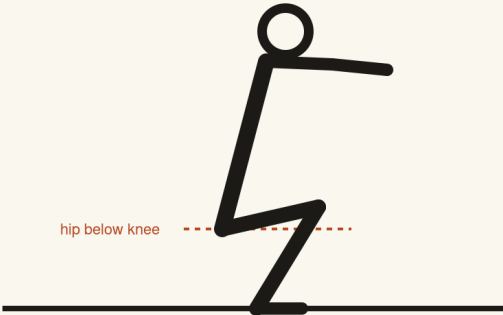
Four capacities carry almost all the mortality signal: **cardiorespiratory fitness, strength, the ability to move/balance, and not sitting all day**. A week that touches all four beats any single "optimal" protocol.

A. Strength — the two patterns that matter most

Strength (and grip), *not muscle mass*, independently predicts mortality — and the dose-response is a **J-curve: benefit peaks around 30–60 minutes per week**, then flattens. Two compound patterns cover most of the body: a **squat** (knee-dominant) and a **hip hinge** (hip-dominant). Add a push, a pull, and a carry and you have a complete program.

Air Squat (bottom)

TIER A · STRENGTH side view — knee-dominant

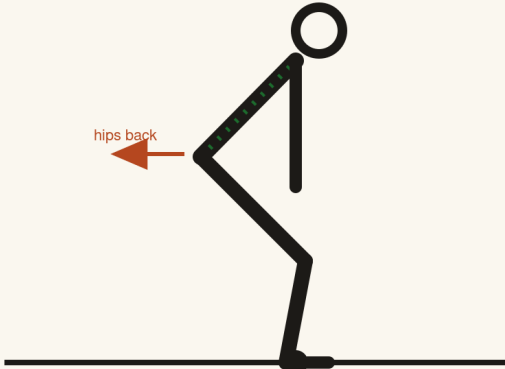


- Hips sit BACK and down; weight on mid-foot
- Hip crease drops below the knee (red line)
- Shins angle forward; heels stay planted
- Chest up, spine long; arms counterbalance

claim: resistance-training-mortality-meta (E) · strength predicts mortality · J-curve ~30-60 min/wk

Hip Hinge / Deadlift

TIER A · STRENGTH side view — hip-dominant



- Push hips BACK (not down); shins near-vertical
- Back stays flat and long (green line)
- Hinge to ~45 deg; arms hang straight to the bar
- Soft knees; weight in the heels

claim: grip-strength-mortality-pure (PURE n=139k) · dexta-strength-not-mass (L)

Air squat (bottom) — knee-dominant, hip below knee

Hip hinge — hips back, flat 45° back, shins vertical

TIER A resistance-training-mortality-meta · grip-strength-mortality-pure (PURE, n=139k) · dexta-strength-not-mass-predicts-mortality. Practical: 2–3 sessions/week, progressive load, the 5 patterns (squat / hinge / push / pull / carry).

B. Cardiorespiratory fitness — the single strongest lever

VO₂max is the strongest mortality predictor in preventive medicine (~5× elite-vs-low, ~13% per MET, **no observed ceiling of benefit**). You raise it two complementary ways, and the evidence does *not* support one being uniquely magic:

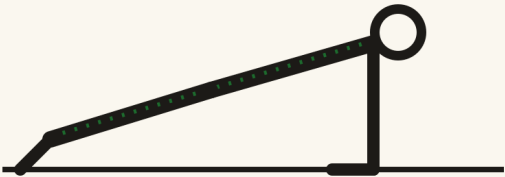
- **Zone 2 (conversational pace), ~150–180 min/week** — builds the aerobic base efficiently. **TIER B CAVEAT** the "uniquely optimal for mitochondria" claim is overstated (conflict-zone2-optimal-mito).
- **VO₂max intervals / HIIT, 1–2× week** — efficiently lifts the ceiling (hiit-crf-cardiometabolic-meta).
- **Just move & break up sitting** — the steepest, *least-confounded* gain is sedentary→active (physical-activity-dose-response-mortality). Steps help, but the benefit **plateaus ~7–8k**, not 10k.

C. Core & balance — the line and the one-leg test

The trunk transmits force in every lift, and a simple balance test independently predicts mortality — and practicing it *is* the intervention. (Two other free tests, the **sit-to-rise** and **chair-rise**, work the same way; real demonstration frames for them sit in [media/](#) for a future photographic library.)

Forearm Plank

TIER A · CORE / STRENGTH *side view — one straight line*

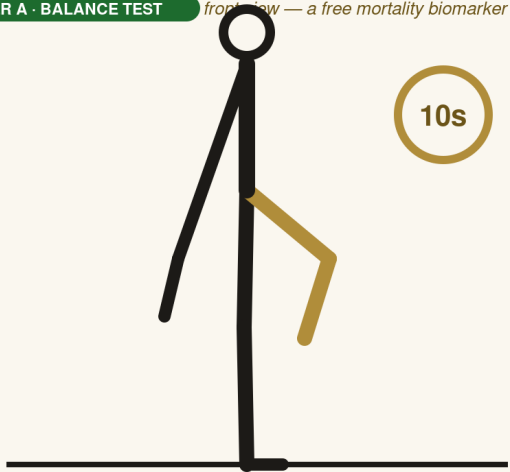


- Ears, hips and ankles on ONE line (green)
- Elbows under shoulders; forearms grounded
- Brace the belly; squeeze glutes; no sag, no pike
- Quality over time: hold only while the line holds

trains the trunk that transmits force in every lift · supports strength + posture

10-Second One-Leg Stand

TIER A · BALANCE TEST *front view — a free mortality biomarker*



- Stand on ONE leg, eyes open, hands off support
- Lift the other knee; hold 10 seconds each side
- Failing it is linked to ~1.84x mortality
- The test IS the training; practice daily

claim: one-leg-stance-10s-mortality · Araujo et al., Br J Sports Med 2022 (HR 1.84)

Forearm plank — one straight braced line

10-second one-leg stand — a free mortality biomarker

TIER A [one-leg-stance-10s-mortality](#) (Araújo 2022, HR 1.84) · plank trains the trunk that protects the spine under load. The balance test is a *predictor* — but the capacity it tests is trainable, so train it directly.

D. Mobility & breath — the supporting tier

Mobility keeps the range you need to keep moving; slow breathing shifts autonomic balance. Both are worth doing; both are honestly Tier B (real acute effects, longevity outcome unproven).

Deep Lunge (hip mobility)

TIER B · MOBILITY

side view — opens the hip flexors

- Front knee stacked over the ankle (~90 deg)
- Back leg long (gold); sink the hips forward
- Torso tall, ribs down; feel the back hip-front
- Counters the shortening from chronic sitting

maps to movement-library/mobility · supports the Tier-A 'move more' lever

Deep lunge — opens the hip flexors; counters sitting

Box Breathing (4-4-4-4)

TIER B · BREATH / HRV

slow breathing shifts autonomic balance

- Inhale 4 · hold 4 · exhale 4 · hold 4
- ~6 breaths/min raises HRV acutely
- Nasal, quiet, diaphragmatic (belly, not chest)
- Real acute effect; longevity outcome unproven

claim: hrv-autonomic-recovery-biomarker (I) · thread-autonomic-hrv · honest tier: surrogate

Box breathing — ~6 breaths/min raises HRV acutely

A simple weekly template

DAY	PRIMARY	CAPACITY TRAINED
Mon	Strength (squat + push + carry)	Strength A
Tue	Zone 2, 45–60 min (walk/bike/row)	CRF base A
Wed	Mobility + balance practice (10 min)	Capacity B
Thu	Strength (hinge + pull)	Strength A
Fri	VO ₂ max intervals (e.g. 4×4 min hard)	CRF ceiling A
Sat	Long easy Zone 2 (hike/ride)	CRF + recovery A
Sun	Rest · sauna · easy walk	Recovery B

Every day: break up sitting, hit the step floor, run through the two balance tests. The template is illustrative — consistency across the four capacities matters more than the exact split.

Part 2 — Lifestyle habits

Outside the gym, a handful of habits carry effect sizes that rival or exceed anything in a supplement stack — and one of them (social connection) is almost never on the list because nothing is sold for it.

HABIT	WHAT THE EVIDENCE SUPPORTS	TIER
Don't smoke	The single largest modifiable mortality factor; the "longevity genes" of GWAS are largely the genetics of not doing harmful things.	A
Protect social connection	Among the largest psychosocial mortality associations — effect size rivals smoking cessation (Holt-Lunstad). Purpose/ikigai adds independently.	A
Sleep ~7 h, regularly	U-shaped mortality with a ~7 h nadir; <i>regularity and timing</i> carry independent weight. Protect it like training.	A
Lower lifetime apoB / LDL	One of the only <i>causal</i> blood levers; cumulative exposure means earlier is better. Diet + (if indicated) medication.	A
Keep metabolic health	Protein-adequate, fiber-rich, mostly-whole-food diet; manage visceral fat & glucose. The pattern matters more than the brand-name diet.	A
Protein adequacy (esp. older adults)	Supports muscle & survival in the elderly — but a genuine mid-life vs late-life tradeoff with mTOR. Not "more is always better".	B
Morning light, dim evenings	Strong circadian mechanism; bright AM / dim PM. (Blue-blocking <i>glasses</i> specifically show no clear benefit — the behavior, not the product.)	B
Early time-restricted eating	Early-window alignment helps insulin/BP — but most of the benefit is the calorie restriction it causes. Late 16:8 is null.	B
Sauna / heat	Dose-response to mortality & dementia — from one Finnish cohort, no RCT. Promising. Infrared ≠ traditional evidence.	B
Cold plunge "for longevity"	Great for mood/alertness; the metabolic <i>outcome</i> data used hours of mild cold, not a 3-min plunge. Enjoy it, don't oversell it.	C

SAFETY — NON-NEGOTIABLE

Never do Wim Hof / breath-hold breathing in or near water (shallow-water blackout drowns people). Cold immersion carries cardiac risk — never combine with breath-holds. Heat/sauna: hydrate, avoid with alcohol, pregnancy, or uncontrolled cardiovascular disease. Loaded spine work: keep it neutral, progress slowly. When in doubt, a clinician beats a protocol.

Test yourself — the high-signal, mostly-free panel

Measurement should be **functional first**. These predict more, and cost less, than the commercial age-tests and continuous monitors.

MEASURE	SIGNAL	TIER
VO ₂ max (hard field test or CPET)	Strongest single mortality predictor	A
Grip · gait speed · chair-rise · 10-s balance · sit-to-rise	Integrated reserve; rivals expensive panels; free	A
apoB (one draw) + Lp(a) (once in life)	The causal blood levers; act on apoB	A
HbA1c + fasting insulin / HOMA-IR	Glycemic control + earliest metabolic warning	A
Overnight HRV · sleep total/timing (wearable)	Within-person <i>trend</i> only — not staging, not cross-device	B
Biological-age clocks · CGM for the healthy		C

Not a validated personal scorecard; clocks disagree; no outcome RCT

The bottom line, in one paragraph

Build and keep **cardiorespiratory fitness** and **strength**; **move more** and sit less; **sleep ~7 h regularly**; keep **apoB/LDL** low across life; eat a protein-adequate, mostly-whole-food pattern; and **protect your relationships**. Train the four capacities — cardio, strength, balance, mobility — and use the free functional tests (VO₂max, grip, gait, sit-to-rise, the 10-second stand) as both training and scorecard. Treat sauna, breathwork, early eating windows and light hygiene as worthwhile *trends*. Treat cold plunges, NAD⁺, glucose monitors for the healthy and biological-age tests as *experiments you enjoy*, not protocols you depend on. The dull list is the powerful list — and it is almost entirely free.

ABOUT THESE ILLUSTRATIONS

The six exercise diagrams are **procedurally generated vector figures** (`reports/gen_exercise_diagrams.py` → SVG → PNG), not AI photos and not stock images. They are reproducible, anatomy-honest (no hallucinated joints), evidence-tagged, and regenerate from code. The corpus also holds **212 real demonstration frames + 29 licensed anatomy images** in `media/` for a future photographic library.

— Optimize Layer · Nucleus Brain · AG Farms · bead bkt-bg6 · not medical advice —